

Benchmarking 1,982 classifier configurations for early severity prediction in acute pancreatitis: an exploratory multi-dataset analysis

Running title: Model-space exploration for acute pancreatitis severity

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Abstract

Large-scale, systematic comparisons across many algorithm families remain uncommon in applied machine learning on small tabular clinical datasets, where most prior studies compare only a handful of models and rarely test whether the

best-ranked model's advantage is statistically distinguishable from noise. We assemble a zoo of 1,982 classifier configurations - linear models, support vector machines, tree ensembles, six independent gradient-boosting implementations, two additional boosting-algorithm variants (DART, GOSS), feedforward and convolutional neural networks, and a fixed-weight hybrid ensemble, including a from-scratch NumPy implementation of classical gradient-boosted TreeBoost that uses none of the standard libraries' internals - and evaluate all of them under one identical 5-fold cross-validation protocol on two independent, publicly available clinical datasets ($n=1,289$ and $n=722$) predicting severe acute pancreatitis. A hybrid DNN+ConvNet+LightGBM ensemble ranks first on the smaller dataset (AUROC=0.857); LightGBM alone remains best on the larger dataset (AUROC=0.889). Using the Hanley-McNeil approximate standard error against a logistic-regression baseline, the top-ranked configuration does not reach conventional statistical significance on the smaller, more imbalanced dataset (one-sided $p=0.054$), while 690 of 1,981 configurations do on the larger dataset - a difference attributable to statistical power, not a qualitative difference between datasets. An engineered, domain-motivated feature (a lab-only organ-dysfunction score inspired by Sepsis-3 methodology) gives mixed, not uniformly positive, results when added to the best model, illustrating that feature-engineering value depends on whether a model already has access to the underlying information. We report all of this - including the negative and mixed findings - honestly, and release the full model-configuration list, benchmark results, and from-scratch reference implementations openly to support replication.

Introduction

Acute pancreatitis (AP) ranges from a mild, self-limited illness to a severe, systemic condition with organ failure and substantial mortality [1–3]. The 2012 Revised Atlanta Classification stratifies AP into mild, moderately severe, and severe categories based on the presence and persistence of organ failure [1], and severe acute pancreatitis (SAP) — persistent organ failure beyond 48 hours — carries mortality estimates ranging roughly from 15% to 40% across published cohorts [1,2,4].

Several validated severity-scoring systems exist — Ranson's criteria, the Bedside Index for Severity in Acute Pancreatitis (BISAP), the Acute Physiology and Chronic Health Evaluation II (APACHE II) score, and the (Modified) CT Severity Index — and have been compared extensively [5–10]. Each has practical limitations for early, automated use: Ranson's criteria require values at admission and again at 48 hours; APACHE II requires more than a dozen variables including several not routinely available at every institution; the CT Severity Index requires imaging typically obtained 72–96 hours after symptom onset [5,6,9]. BISAP was designed to be calculable at admission from a small number of routine variables and performs comparably to APACHE II in several studies, though findings vary by cohort [5,6,7,8].

Supervised machine learning has increasingly been applied to SAP prediction using routinely collected admission laboratory values [11–18]. Reported studies span logistic regression, random forests, gradient-boosted decision trees (GBDT), and deep learning, including convolutional networks applied to CT imaging

[16,17]. Reported discrimination is frequently high (AUROC often exceeding 0.90), though external validation on independent institutions or populations is uncommon [12,13,14]. A 2024 systematic review and meta-analysis of machine-learning-based AP severity studies concluded that such models generally match or modestly exceed traditional scoring systems, while noting substantial between-study heterogeneity in cohort definitions and outcome operationalization [19].

Most prior comparative studies evaluate a small number of algorithm families (commonly logistic regression, one tree ensemble, and one boosting or deep-learning model). We are not aware of a prior study that benchmarks a comparably large model space — spanning classical linear models through six independent gradient-boosting implementations, two additional boosting-algorithm variants, deep feedforward and convolutional architectures, and a hybrid ensemble — under one identical cross-validation protocol on two independent public datasets, together with formal statistical-significance testing against a common baseline. The present study addresses this gap. We additionally test whether an engineered, lab-only organ-dysfunction score inspired by the Sepsis-3 Sequential Organ Failure Assessment (SOFA) framework [20], built using only its scoring methodology (not any sepsis patient data), improves the best-performing model.

Methods

Datasets. Two de-identified, publicly released tabular datasets were used, both reportedly originating from the Second Affiliated Hospital of Guilin Medical University: “ap_multiml” (n=1,289; 204 SAP cases, 15.8%; 59 features after

sanitization; MIT license) [21] and “ap_lnn” (n=722; 137 SAP cases, 19.0%; 106 features after sanitization; Apache-2.0 license) [22]. Both consist exclusively of routine admission laboratory values (complete blood count, coagulation panel, liver and renal function, electrolytes, inflammatory markers, and, in ap_lnn, lactate) plus age and sex; neither includes vital signs, imaging, or Glasgow Coma Scale. Direct patient identifiers were absent from the released files; no re-identification was attempted. During sanitization we identified and corrected a label-direction inconsistency in the source files (the majority, non-SAP class was coded 1 in both datasets) that would have affected all threshold-dependent metrics (F1, sensitivity, specificity) had it not been corrected; area-under-the-ROC-curve (AUROC) values are invariant to a joint label/score flip and were unaffected.

Model zoo. We programmatically assembled 1,982 classifier configurations spanning: logistic regression, ridge classification, stochastic gradient descent (multiple losses/penalties), perceptron, linear and kernel support vector machines; k-nearest neighbors; Gaussian and Bernoulli naive Bayes; decision trees and extremely randomized trees (including unconstrained-leaf variants); random forests [23] and extremely randomized tree ensembles [24]; AdaBoost [25]; bagging; scikit-learn's GradientBoostingClassifier (classical TreeBoost [26]) and HistGradientBoostingClassifier; XGBoost [27]; LightGBM [28]; CatBoost [29]; a from-scratch NumPy implementation of Friedman-style TreeBoost, using none of the above libraries' internals; XGBoost's DART (dropout-tree) booster [30]; LightGBM's DART and GOSS boosting modes [28,30]; CatBoost's Plain (non-ordered) boosting mode [29]; feedforward and one-dimensional convolutional neural networks implemented in PyTorch [31], using batch normalization [32], dropout [33], and the Adam optimizer [34]; and a

hybrid ensemble combining a feedforward network, a convolutional network, and LightGBM via one of three fixed weighting schemes. All scikit-learn-family models used scikit-learn [35]. Every configuration used an identical preprocessing pipeline (median imputation, standardization) and identical 5-fold stratified cross-validation [36], with all reported metrics computed on out-of-fold predictions. Class imbalance (15.8%/19.0% SAP prevalence) was handled per-model via built-in class-weighting options where available and via custom class-balanced gradient weighting in the from-scratch implementations; oversampling methods such as SMOTE [37] were not used.

Statistical significance testing. Each model's AUROC was compared against a single, fixed logistic-regression baseline using the Hanley–McNeil approximate standard error for AUROC [38] and a one-sided, unpaired z-test (testing whether the candidate's AUROC significantly exceeds the baseline's). This approach is conservative: because every model was evaluated on the same cross-validation folds, a properly paired test such as DeLong's method [39] would generally have higher power to detect differences than the unpaired approximation used here; a paired analysis was not performed because per-model out-of-fold prediction vectors were not retained for all 1,982 configurations during benchmarking.

Engineered feature: a lab-only quasi-SOFA score. Motivated by the Sepsis-3 SOFA framework's organ-dysfunction scoring methodology [20] — used here only as a scoring methodology, not with any sepsis patient data — we constructed a lab-only “quasi-SOFA” score summing renal (creatinine), hepatic (bilirubin), and coagulation (platelet count) sub-scores using SOFA's threshold bands, with optional additive terms for leukocyte abnormality and, where available, lactate above 2 mmol/L. This is not a validated SOFA score: full SOFA

additionally requires respiratory, cardiovascular, and neurological components not available in either dataset used here. We evaluated quasi-SOFA both as a standalone predictor and as an added feature to the best-performing hybrid model.

Software and reproducibility. All models used a fixed random seed. Analysis code, the full model-configuration list, and benchmark checkpoints are openly available (Data Availability Statement).

Results

Model-space growth and overall ranking. The compared model space grew from an initial 171 gradient-boosting-focused configurations to 1,982, across five expansion stages (linear/SVM/KNN/tree/ensemble/GBDT hyperparameter expansion; DART/GOSS/Plain boosting and a from-scratch GBDT; a learning-rate grid extension; unconstrained-leaf “huge-tree” variants; and deep-learning/hybrid architectures). Of 1,982 configurations, 1,981 were evaluated successfully on both datasets; the sole consistent failure was quadratic discriminant analysis, which raised a collinearity error on both datasets.

On `ap_multiml`, a hybrid ensemble configuration (feedforward network + one-dimensional convolutional network + LightGBM, GBDT-heavy weighting) achieved the highest AUROC of all 1,981 evaluated configurations (0.8566), narrowly ahead of the best single-library LightGBM configuration (0.8537, rank 2) and ahead of the three headline single-library GBDT configurations used in

our earlier three-library comparison (LightGBM rank 102/1,981, AUROC=0.8499; XGBoost rank 506/1,981, AUROC=0.8421; CatBoost rank 1,020/1,981, AUROC=0.8290). On `ap_lnn`, the best-ranked configuration remained a LightGBM model (AUROC=0.8892), with the best hybrid ensemble ranked 168th (AUROC=0.8799).

New boosting-algorithm variants. XGBoost's DART booster, LightGBM's DART and GOSS modes, and CatBoost's Plain boosting mode each performed comparably to, but did not exceed, their respective standard-boosting counterparts on either dataset (best AUROC range 0.836–0.847 on `ap_multiml` and 0.866–0.885 on `ap_lnn`). Neither the feedforward network nor the convolutional network alone matched the best gradient-boosted tree ensembles, consistent with prior reports that tree ensembles remain competitive with, and often superior to, deep learning on small-to-moderate tabular datasets [40]; the fixed-weight hybrid combination, however, outperformed every individual family on `ap_multiml`.

Statistical significance against a logistic-regression baseline. Using the unpaired Hanley–McNeil approach (baseline AUROC=0.8158 on `ap_multiml`, 0.8087 on `ap_lnn`), zero of 1,981 configurations reached $p < 0.05$ on `ap_multiml` — including the top-ranked hybrid model itself (one-sided $p = 0.054$). At $p < 0.10$, 154 configurations reached significance on `ap_multiml`. On `ap_lnn`, 690 of 1,981 configurations reached $p < 0.05$, including the top-ranked LightGBM configuration (one-sided $p \approx 0.0037$). This asymmetry reflects statistical power: with only 204 SAP cases in `ap_multiml`, an AUROC gap of 0.041 over baseline is difficult to distinguish from noise under this conservative unpaired test, even though a comparable or larger absolute gap is readily distinguished on `ap_lnn`.

Sensitivity, specificity, and false-negative trade-offs. Among statistically significant models, a random-forest configuration achieved both the lowest false-negative count and, simultaneously, the highest F1 score within that low-false-negative group on each dataset (ap_multiml: F1=0.524, sensitivity=46.1%, 110/204 false negatives; ap_lnn: F1=0.631, sensitivity=68.6%, specificity=88.5%, 43/137 false negatives). Separately, one deep-learning configuration on ap_multiml reached 75.5% sensitivity (50/204 false negatives) at a nearly identical F1 (0.518) to the random-forest configuration, illustrating that F1 alone can obscure clinically relevant differences in the sensitivity/specificity balance.

Quasi-SOFA feature. As a standalone predictor, quasi-SOFA reached AUROC 0.663 (ap_multiml) and 0.678 (ap_lnn). Adding it as a feature to the best hybrid ensemble produced mixed results: AUROC and F1 both decreased slightly on ap_multiml (0.857 $\rightarrow$ 0.848; 0.547 $\rightarrow$ 0.540), while on ap_lnn AUROC decreased slightly (0.880 $\rightarrow$ 0.878) but F1 increased (0.607 $\rightarrow$ 0.620) alongside a five-case reduction in false negatives (62 $\rightarrow$ 57).

Discussion

Two findings stand out. First, expanding the compared model space by roughly an order of magnitude beyond a typical three-library gradient-boosting comparison did not substantially change the qualitative conclusion: gradient-boosted tree ensembles and closely related methods (random forests, extremely

randomized trees) occupy essentially all top-ranked positions on both datasets, and the single best result found across the entire space — the hybrid ensemble on `ap_multiml` — exceeded the best pure-GBDT result by only 0.003 AUROC, a difference that itself does not reach conventional statistical significance against even a simple baseline. This is consistent with reports that deep learning does not reliably surpass gradient boosting on small-to-moderate tabular datasets [40], and suggests that further architecture search alone is unlikely to yield a qualitatively different result on datasets of this size without additional data.

Second, the quasi-SOFA feature-engineering result illustrates a common, easily overlooked pitfall: a feature motivated by domain knowledge from an adjacent clinical field (sepsis) did not reliably improve a model that already had direct access to the same underlying raw laboratory values used to construct that feature. A plausible explanation is that the hybrid model's gradient-boosted component can already learn comparable interaction terms directly from creatinine, bilirubin, and platelet count; the hand-crafted composite therefore adds little information the model could not otherwise learn. We report this as a genuine negative/mixed result rather than omitting it.

This study has several limitations. It is a secondary analysis of two publicly released datasets from a single reporting institution; no data were collected prospectively by the authors, and no external validation on an independent institution or population was performed. The significance testing used an unpaired approximation; a properly paired test (DeLong [39]) evaluated on the same cross-validation folds would likely have higher power and was not performed because per-model out-of-fold prediction vectors were not retained during large-scale benchmarking. The quasi-SOFA score is a research

approximation that omits SOFA's respiratory, cardiovascular, and neurological components, none of which are available in either dataset. Evaluating 1,982 configurations on the same folds within each dataset increases the risk that the single best-ranked configuration reflects some degree of selection on noise, which the significance testing was specifically intended to address (and is precisely why the top-ranked hybrid model's failure to reach $p < 0.05$ is a meaningful, not merely technical, finding). Model calibration and decision-curve analysis were not performed across the full model zoo; all comparisons here concern discrimination only, which is necessary but not sufficient to establish clinical usefulness. Finally, this report inherits the limitations of any retrospective, non-externally-validated model-development study: it is not validated for clinical use and should not inform patient care.

Conclusion

Across a model space roughly an order of magnitude larger than typically reported in this literature, no single algorithm family dominates; a hybrid, hand-weighted combination of gradient boosting and deep learning achieved the best rank on one dataset without reaching conventional statistical significance against a simple baseline, and an engineered organ-dysfunction feature did not consistently improve performance. These findings do not change the fundamental limitations of secondary analysis on small, single-institution public datasets: external validation, prospective evaluation, and clinical/regulatory review remain prerequisites for any clinical use. We release all code, model configurations, and benchmark results openly to support replication and extension by others.

Data availability statement

The two datasets analyzed in this study are third-party datasets originally released by their respective authors and are publicly available at: <https://github.com/longshike/Predicting-acute-pancreatitis-severity-with-multi-machine-learning-models> (MIT license; “ap_multiml”) and <https://github.com/longshike/LNN-for-SAP-Prediction> (Apache-2.0 license; “ap_lnn”). Sanitized/translated derivative versions used in this analysis, all model-configuration code, the full 1,982-configuration benchmark results, and the from-scratch reference implementations described in the Methods are available at <https://github.com/netanelcyber/penuX> (PenuX-AP-Severity subdirectory) under an open-source license.

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Competing interests

The author declares no competing interests.

Author contributions

N.S.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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